Python Core for AI/ML

Everything in Python you need before touching ML libraries - from installation to OOP, files, decorators, async and testing.

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1. Installing Python, venv, pip, conda, Jupyter, VS Code [Beginner]

Explanation (English)
Install Python 3.11+. Use a virtual environment per project (python -m venv) so package versions don't clash. pip installs packages; conda (Miniconda) manages envs and binary packages. Jupyter/Colab give interactive notebooks.

Tanglish (Tamil + English) Vilakkam
Python 3.11+ install pannunga. Ovvoru project-kum thani virtual environment use pannunga. pip package install, conda env manage. Jupyter/Google Colab notebook-ku.

Example / Program
```
python -m venv .venv
source .venv/bin/activate   # Windows: .venv\Scripts\activate
pip install numpy pandas jupyter
pip freeze > requirements.txt
jupyter notebook
```

Learn more
YouTube search: python installation venv jupyter vscode setup

2. Variables, data types, operators, type casting [Beginner]

Explanation (English)
int, float, str, bool, None; arithmetic, comparison, logical operators. Python is dynamically typed. Use type() and int()/float()/str() to convert.

Tanglish (Tamil + English) Vilakkam
Variable-ku type declare panna venam. int, float, str, bool. type() check, int()/float() convert.

Example / Program
```
x=10;y=3.5;name='AI';ok=True
print(type(x),x/3,x//3,x%3,x**2,int(y),str(x)+name)
```

Learn more
YouTube search: Variables, data types, operators, type casting tutorial

3. Strings and f-strings [Beginner]

Explanation (English)
Strings are immutable sequences; slicing, split/join/strip/replace, formatting with f-strings and format().

Tanglish (Tamil + English) Vilakkam
String immutable. slicing, split, join, f-string.

Example / Program
```
s='Machine Learning'
print(s[:7],s.lower(),s.split(),f'{s!r:>25}',f'{3.14159:.2f}')
```

Learn more
YouTube search: Strings and f-strings tutorial

4. Control flow: if/elif/else, loops, break/continue, match [Beginner]

Explanation (English)
Branching with if/elif/else, loops with for/while, range(), enumerate(), zip(); structural pattern matching with match/case (3.10+).

Tanglish (Tamil + English) Vilakkam
if/else, for, while loop. enumerate, zip romba useful.

Example / Program
```
for i,(a,b) in enumerate(zip('abc',[1,2,3])):
    if b==2: continue
    print(i,a,b)
```

Learn more
YouTube search: Control flow: if/elif/else, loops, break/continue, match tutorial

5. Lists, tuples, sets, dictionaries [Beginner]

Explanation (English)

list (ordered mutable), tuple (ordered immutable), set (unique, fast membership), dict (key-value hash map). Know their methods and Big-O.

Tanglish (Tamil + English) Vilakkam

list mutable, tuple immutable, set unique values, dict key-value.

Example / Program

```
l=[3,1,2];l.sort();t=(1,2);s={1,2,2};d={'a':1}
d['b']=2;print(l,t,s,d.get('c',0),list(d.items()))
```

Learn more

YouTube search: Lists, tuples, sets, dictionaries tutorial

6. Comprehensions and generators [Intermediate]

Explanation (English)

List/dict/set comprehensions create collections in one line. Generators (yield / generator expressions) produce items lazily and save memory - vital for big data.

Tanglish (Tamil + English) Vilakkam

Comprehension short-a list create pannum. Generator memory save pannum (lazy).

Example / Program

```
sq=[x*x for x in range(5) if x%2==0]
d={k:len(k) for k in ['ai','ml']}
g=(x*x for x in range(10**9))
print(sq,d,next(g))
```

Learn more

YouTube search: Comprehensions and generators tutorial

7. Functions: args, kwargs, defaults, lambda, scope, recursion [Beginner]

Explanation (English)

def with positional/keyword/default args, *args and **kwargs, lambda one-liners, LEGB scope, recursion. Avoid mutable default arguments.

Tanglish (Tamil + English) Vilakkam

def function, *args, **kwargs, lambda. Mutable default argument use pannadheenga.

Example / Program

```
def f(a,b=2,*args,**kw): return a+b+sum(args)
print(f(1),f(1,2,3,4,x=5),(lambda x:x*2)(4))
```

Learn more

YouTube search: Functions: args, kwargs, defaults, lambda, scope, recursion tutorial

8. map, filter, reduce, sorted, any, all, zip [Intermediate]

Explanation (English)

Functional helpers for transforming iterables. sorted(key=...) is used everywhere for ranking.

Tanglish (Tamil + English) Vilakkam

map/filter/sorted - data transform panna quick-a use pannalam.

Example / Program

```
from functools import reduce
print(list(map(str,[1,2])),list(filter(None,[0,1,2])),reduce(lambda a,b:a*b,[1,2,3,4]),sorted(['bb','a'],key=len))
```

Learn more

YouTube search: map, filter, reduce, sorted, any, all, zip tutorial

9. Modules, packages, imports, __name__ == '__main__' [Beginner]

Explanation (English)

A module is a .py file; a package is a folder with __init__.py. Use import, from-import, aliases. The __main__ guard prevents code from running on import.

Tanglish (Tamil + English) Vilakkam

Oru .py file = module. Folder + __init__.py = package. __main__ guard useful.

Example / Program

```
import math as m
from collections import Counter
if __name__=='__main__':
    print(m.sqrt(16),Counter('banana'))
```

Learn more

YouTube search: Modules, packages, imports, __name__ == '__main__' tutorial

10. File handling: text, CSV, JSON, pickle, pathlib [Beginner]

Explanation (English)

Use with-open for safe file handling; csv and json modules; pathlib.Path for paths; pickle/joblib to save Python objects (models).

Tanglish (Tamil + English) Vilakkam

with open() use pannunga, auto close aagum. JSON, CSV, pickle, pathlib.

Example / Program

```
import json,pathlib
p=pathlib.Path('a.json');p.write_text(json.dumps({'x':1}))
print(json.loads(p.read_text()))
```

Learn more

YouTube search: File handling: text, CSV, JSON, pickle, pathlib tutorial

11. Exception handling [Beginner]

Explanation (English)

try/except/else/finally, raising exceptions, custom exceptions. Catch specific exceptions, never a bare except.

Tanglish (Tamil + English) Vilakkam

try/except use panni error handle pannunga. Specific exception catch pannunga.

Example / Program

```
try:
    1/0
except ZeroDivisionError as e:
    print('err',e)
finally:
    print('done')
```

Learn more

YouTube search: Exception handling tutorial

12. Object-Oriented Programming: classes, inheritance, dunder methods [Intermediate]

Explanation (English)

Classes bundle data and behaviour. Inheritance, polymorphism, encapsulation, @property, @classmethod, @staticmethod, __init__, __repr__, __call__. PyTorch models and sklearn estimators are classes.

Tanglish (Tamil + English) Vilakkam

Class = blueprint, object = instance. Inheritance, polymorphism. PyTorch model, sklearn estimator ellaam class dhaan.

Example / Program

```
class Model:
    def __init__(self,w): self.w=w
    def __call__(self,x): return self.w*x
```

```
    def __repr__(self): return f'Model({self.w})'
print(Model(2)(5),Model(2))
```

Learn more
YouTube search: Object-Oriented Programming: classes, inheritance, dunder methods tutorial

13. Dataclasses, typing and type hints [Intermediate]

Explanation (English)
dataclasses reduce boilerplate; type hints (list[int], Optional, Protocol) improve readability and enable mypy/pyright checks.

Tanglish (Tamil + English) Vilakkam
dataclass boilerplate kuraikkum. Type hints code-a clean-a vaikkum.

Example / Program
```
from dataclasses import dataclass
@dataclass
class Cfg:
    lr: float=1e-3
    epochs: int=10
print(Cfg(epochs=5))
```

Learn more
YouTube search: Dataclasses, typing and type hints tutorial

14. Decorators and context managers [Advanced]

Explanation (English)
Decorators wrap functions to add behaviour (timing, caching, logging). Context managers (with) manage setup/teardown (files, GPU contexts, torch.no_grad()).

Tanglish (Tamil + English) Vilakkam
Decorator function-a wrap panni extra behaviour kudukkum. with statement = context manager.

Example / Program
```
import time,functools
def timer(f):
    @functools.wraps(f)
    def w(*a,**k):
        t=time.time();r=f(*a,**k);print(time.time()-t);return r
    return w
@timer
def run(): return sum(range(10**6))
run()
```

Learn more
YouTube search: Decorators and context managers tutorial

15. Iterators, iterables and itertools [Advanced]

Explanation (English)
An iterator implements __iter__/__next__. itertools provides product, permutations, chain, groupby, islice for efficient loops (grid search combos).

Tanglish (Tamil + English) Vilakkam
Iterator __next__ implement pannum. itertools combos generate panna useful.

Example / Program
```
import itertools as it
print(list(it.product([0.1,0.01],[16,32])))
```

Learn more
YouTube search: Iterators, iterables and itertools tutorial

16. collections: Counter, defaultdict, deque, namedtuple, heapq [Intermediate]

Explanation (English)
Specialised containers: Counter for frequencies, defaultdict for grouping, deque for queues, heapq for top-k.

Tanglish (Tamil + English) Vilakkam
Counter frequency count, defaultdict grouping, heapq top-k.

Example / Program
```
from collections import Counter,defaultdict
import heapq
print(Counter('mississippi').most_common(2),heapq.nlargest(2,[5,1,9,3]))
```

Learn more
YouTube search: collections: Counter, defaultdict, deque, namedtuple, heapq tutorial

17. Standard library essentials: os, sys, re, datetime, random, math, logging, argparse [Intermediate]

Explanation (English)
os/sys for system, re for regex, datetime for time, random/math, logging instead of print, argparse for CLI scripts and training scripts.

Tanglish (Tamil + English) Vilakkam
os, sys, re (regex), datetime, logging, argparse - daily use libraries.

Example / Program
```
import re,datetime,logging
print(re.findall(r'\d+','a12b345'),datetime.date.today())
logging.basicConfig(level=logging.INFO);logging.info('hi')
```

Learn more
YouTube search: Standard library essentials: os, sys, re, datetime, random, math, logging, argparse tutorial

18. Regular expressions [Intermediate]

Explanation (English)
Pattern matching for text cleaning in NLP: search, findall, sub, groups, greedy vs lazy matching.

Tanglish (Tamil + English) Vilakkam
Text clean panna regex romba use. findall, sub, groups.

Example / Program
```
import re
print(re.sub(r'[^a-z ]','','Hello, World! 123'.lower()))
```

Learn more
YouTube search: Regular expressions tutorial

19. Algorithms and data structures for ML engineers [Intermediate]

Explanation (English)
Big-O, sorting, searching, hash maps, stacks/queues, trees, graphs, DP. Vectorise instead of looping for speed.

Tanglish (Tamil + English) Vilakkam
Big-O, sorting, searching, tree, graph, DP basics. Loop-a vida vectorization fast.

Example / Program
```
def bsearch(a,x):
    lo,hi=0,len(a)-1
    while lo<=hi:
        m=(lo+hi)//2
        if a[m]==x: return m
        lo,hi=(m+1,hi) if a[m]<x else (lo,m-1)
    return -1
print(bsearch([1,3,5,7],5))
```

Learn more

YouTube search: Algorithms and data structures for ML engineers tutorial

20. Concurrency: threading, multiprocessing, asyncio, the GIL [Advanced]

Explanation (English)
The GIL limits CPU threads; use multiprocessing for CPU-bound, threads/asyncio for I/O-bound work. DataLoader num_workers uses processes.

Tanglish (Tamil + English) Vilakkam
GIL-nala CPU work-ku multiprocessing, I/O-ku threading/asyncio.

Example / Program
```
from concurrent.futures import ProcessPoolExecutor
def sq(x): return x*x
if __name__=='__main__':
    with ProcessPoolExecutor() as ex: print(list(ex.map(sq,range(5))))
```

Learn more
YouTube search: Concurrency: threading, multiprocessing, asyncio, the GIL tutorial

21. Performance: profiling, numba, cython, caching, memory [Advanced]

Explanation (English)
Measure first (cProfile, timeit, line_profiler); then vectorise (NumPy), cache (functools.lru_cache), JIT (numba), or use C extensions.

Tanglish (Tamil + English) Vilakkam
Mudhalla profile pannunga. Appuram vectorize, lru_cache, numba.

Example / Program
```
import functools
@functools.lru_cache(None)
def fib(n): return n if n<2 else fib(n-1)+fib(n-2)
print(fib(80))
```

Learn more
YouTube search: Performance: profiling, numba, cython, caching, memory tutorial

22. Testing and code quality: pytest, black, ruff, mypy, pre-commit [Intermediate]

Explanation (English)
Write unit tests with pytest, format with black/ruff, type-check with mypy. Tests protect data pipelines and model code from silent bugs.

Tanglish (Tamil + English) Vilakkam
pytest test, black/ruff format, mypy type check. ML code-ku test romba mukkiyam.

Example / Program
```
# test_add.py
def add(a,b): return a+b
def test_add(): assert add(1,2)==3
# run: pytest -q
```

Learn more
YouTube search: Testing and code quality: pytest, black, ruff, mypy, pre-commit tutorial

23. Git and GitHub for ML projects [Beginner]

Explanation (English)
init, add, commit, branch, merge, pull request; .gitignore for data/models; use Git LFS or DVC for large files.

Tanglish (Tamil + English) Vilakkam
git init, add, commit, push. Big data/model files git-la podadheenga - DVC/LFS use pannunga.

Example / Program
```
git init
```

```
git add .
git commit -m 'first commit'
git branch -M main
git remote add origin <url>
git push -u origin main
```

Learn more

YouTube search: Git and GitHub for ML projects tutorial

24. Working with APIs and web scraping (requests, BeautifulSoup) [Intermediate]

Explanation (English)

Collect data using REST APIs (requests, JSON) and scrape HTML responsibly (robots.txt) with BeautifulSoup / Scrapy / Selenium.

Tanglish (Tamil + English) Vilakkam

requests library API call, BeautifulSoup scraping. robots.txt respect pannunga.

Example / Program

```
import requests
r=requests.get('https://api.github.com');print(r.status_code,list(r.json())[:2])
```

Learn more

YouTube search: Working with APIs and web scraping (requests, BeautifulSoup) tutorial
Docs: https://requests.readthedocs.io

25. SQL basics for data scientists (SQLite, joins, group by, window functions) [Intermediate]

Explanation (English)

Query relational data: SELECT, WHERE, JOIN, GROUP BY, HAVING, CTEs, window functions. Use sqlite3 or pandas.read_sql.

Tanglish (Tamil + English) Vilakkam

SELECT, JOIN, GROUP BY, window function. Data scientist-ku SQL must.

Example / Program

```
import sqlite3
c=sqlite3.connect(':memory:')
c.execute('create table t(a int)');c.executemany('insert into t values(?)',[(1,),(2,)])
print(c.execute('select sum(a) from t').fetchall())
```

Learn more

YouTube search: SQL basics for data scientists (SQLite, joins, group by, window functions) tutorial

26. Command line, Linux and shell basics [Beginner]

Explanation (English)

cd, ls, grep, find, pipes, chmod, ssh, tmux, nvidia-smi. Cloud GPUs and servers are used via the terminal.

Tanglish (Tamil + English) Vilakkam

Terminal commands: cd, ls, grep, ssh. Cloud GPU use panna venum.

Example / Program

```
ls -la | grep .py
nvidia-smi
tmux new -s train
```

Learn more

YouTube search: Command line, Linux and shell basics tutorial

27. Python practice roadmap: 100 beginner exercises to projects [Beginner]

Explanation (English)

Practise on HackerRank, LeetCode (easy), Exercism, Advent of Code. Build: calculator, to-do CLI, expense tracker, web scraper, data dashboard.

Tanglish (Tamil + English) Vilakkam

HackerRank, LeetCode easy, Exercism-la practice. Calculator, todo app, scraper project pannunga.

Learn more

YouTube search: python projects for beginners
Exercism: https://exercism.org/tracks/python
Python docs tutorial: https://docs.python.org/3/tutorial/